AI-Based Electricity Demand Intelligence System Sri Lanka

Project ID

R26-IT010

Project Proposal Report

Madushani R.P– IT22273062

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

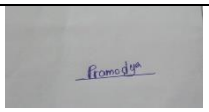
Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Madushani R P	**IT22273062**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Nathalida Silva	20/03/2026	

Abstract

Forecasting electricity usage is critical for the effective planning of energy resources and the management of Smart Grid systems. Many conventional ML modelling methodologies (e.g., Random Forest, XGBoost, and other deep learning architectures) produce highly accurate forecasts yet provide little insight into the reasons behind demand volatility as they are treated as black boxes. The inability of these modelling methods to provide insightful content into operating parameters leads to the mistrust of the model and subsequently to their limited practical implementation, especially during periods of peak demand or when demand is affected by some anomaly.

To resolve this issue, we are developing an Explainable Artificial Intelligence (XAI) component to be integrated into a demand forecasting system that will allow for an explanation of the forecast through an analysis of features associated with demand forecasting. These features include weather conditions (e.g. temperature, humidity, and rainfall), the time of day, the day of the week, the month, any public events that could have a positive or negative effect on demand, and the historical demand for each respective hour of the day.

Numerous XAI methodologies (e.g. SHAP (SHapley Additive Explanations), LIME, and Permutation Feature Importance) will be utilized to provide global and local explanations of the forecasts generated by the demand forecasting and to provide special analyses of the forecasts for peak, low, anomaly, and public event days through the use of visual dashboards and summary (bubble) plots.

Our goal for the demand forecasting system is to provide transparency, trust, and informed decision-making with respect to optimizing load management, improving energy efficiency, and decreasing the risks associated with blackouts.

Keywords: Explainable AI, Electricity Demand Forecasting, SHAP, XGBoost, Feature Importance

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project supervisor, **Dr. Nathali Silva**, for her valuable guidance, continuous support, and constructive feedback throughout the development of this research project. Her expertise and encouragement greatly helped us in understanding the research problem and improving the quality of our work.

I am also thankful to the Department of Information Technology and the Sri Lanka Institute of Information Technology (SLIIT) for providing the academic environment, learning resources, and technical support required to successfully carry out this research.

My heartfelt appreciation goes to my project team members — Thanujan T (IT22545176), Janahan R (IT22069740), and Dewmini D.M.D.P (IT22630216) — for their cooperation, dedication, and teamwork throughout the project. Their collaborative efforts, idea sharing, and commitment played an important role in the successful progress of this research.

Finally, I would like to thank my family and friends for their continuous encouragement, understanding, and moral support during the completion of this research project.

TABLE OF CONTENT

DECLARATION 2
Abstract 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENT 5
LIST OF TABLES 6
LIST OF FIGURES 6
LIST OF ABBREVIATIONS 7
LIST OF APPENDICES 8
1.INTRODUCTION 9
1.1 Background and Context 9
1.2 Literature Review 10
1.2.1 Electricity Demand Forecasting and the Sri Lankan Context 10
1.2.2 Machine Learning Approaches for Load Forecasting 10
1.2.3 Deep Learning Models for Time-Series Forecasting 10
1.2.4 Explainable Artificial Intelligence (XAI) in Energy Systems 10
1.2.5 Visualization and Decision Support Tools 11
1.2.6 Limitations of Existing Systems 11
1.3 Research Gap 11
Key Gaps: 11
Technical Gaps: 12
2.OBJECTIVE 13
2.1 Main Objective 13
2.2 Specific Objectives 13
3.METHODOLOGY 14
3.1.Input Data and Pre-Trained Forecast Model ..... Error! Bookmark not defined.
3.2.Explainability Methods 14
3.3.Global and Local Explanations 14
3.4.Special Case Analysis 14
3.5.Visualization and Dashboard Development 15
3.6.Backend and Frontend Implementation 15
3.7.Validation Strategy 15
3.8.Technology Stack 19

4.HIGH-LEVEL SYSTEM ARCHITECTURE 20
4.1 Module Overview 20
4.2 Data Flow 20
4.3 Backend Architecture 20
4.4 Integration with Other Components 21
4.5 Scalability and Privacy 21
5.USER REQUIREMENTS 22
5.1 Stakeholder Identification 22
5.2 Functional Requirements 22
5.3 Non-Functional Requirements 23
6.COMMERCIALIZATION PLAN 24
7.BUDGET AND JUSTIFICATION 25
8.WORK BREAKDOWN STRUCTURE (WBS) 27
9.GANTT CHART 29
10.References List 30
11.APPENDICES 31

LIST OF TABLES

Table 1:Technology Stack Summary 19
Table 2 :Functional Requirements 22
Table 3 :Non Functional Requirements 23
Table 4: Budget and Budget Justification 25
Table 5:work breakdown structure 27
Table 6:Gantt Chart 29
Table 7:Risk Analysis 31

LIST OF FIGURES

Figure 0-1:System Architecture Diagram 16
Figure 0-2:Component Architecture Diagram 17
Figure 0-3:Research Methonology Diagram 18

LIST OF ABBREVIATIONS

AI - Artificial Intelligence

XAI - Explainable Artificial Intelligence

SHAP - SHapley Additive Explanations

LIME - Local Interpretable Model-Agnostic Explanations

ARIMA - AutoRegressive Integrated Moving Average

SARIMA - Seasonal AutoRegressive Integrated Moving Average

LSTM - Long Short-Term Memory

GRU - Gated Recurrent Unit

ML - Machine Learning

CSV - Comma-Separated Values

KPI - Key Performance Indicator

PDF - Portable Document Format

API - Application Programming Interface

GUI - Graphical User Interface

IoT - Internet of Things

DBSCAN - Density-Based Spatial Clustering of Applications with Noise

KMeans - Clustering Algorithm

LIST OF APPENDICES

Appendix A: Risk Analysis

Appendix B: Ethical Considerations and Data Collection Consent Framework

Appendix C: Technology and Framework Stack

Appendix D: Preliminary UI Mockups – Translation Screen

1.INTRODUCTION

1.1 Background and Context

Electricity supply is very essential for socio-economic growth in any country. The electricity demand in Sri Lanka is changing very rapidly due to various reasons, such as weather changes, industrial activities, and holidays. The changing electricity demand is a major problem faced by electricity boards and industries, resulting in excess or insufficient supply, higher operating costs, power outages, and inefficient use of electricity. The electricity supply is directly linked to people, industries, and socio-economic growth..

In the global community, the use of electricity demand forecasting is now an integral part of the management of energy. For example, various research projects have been conducted in Nigeria and elsewhere using artificial intelligence (AI) to forecast the energy demands of the future more accurately while providing meaningful insights into the same [1]. In another example, the use of spatiotemporal decoupling coupled with explainable AI has been explored for the purposes of predicting power loads and prices in an industrial setting, thereby showcasing the global move toward the incorporation of AI for energy intelligence [2]. However, the conventional approach is mostly based on predicting the energy demands without explaining the underlying causal factors of the same [3][4][5].

The solutions that are currently in use in Sri Lanka are based on statistical forecasting models. These models can only make short-term predictions but do not offer detailed information on the reasons behind the variations in demand. Such solutions are not transparent, behavioral, or scenario-based. This makes it difficult for the planners to understand the factors that influence the demand, such as temperature, holidays, or changes in industrial load, and take proactive decisions to manage risks.

To overcome these challenges, the author of this research proposes an Explainable and Causal AI System as part of an integrated Electricity Demand Intelligence System. In the proposed system, there will be analysis of the contribution of factors to the demand for electricity by applying SHAP and LIME techniques. There will be generation of day-wise reports on the cause-effect analysis of the demand for electricity. This will help in understanding the demand for electricity in terms of behavioral factors as well as the causes of the demand. In addition, there will be the use of scenario-based simulation techniques in the proposed system to perform “what-if” analysis of the potential impact of hypothetical scenarios on the demand for electricity.

This research aligns with the United Nations' Sustainable Development Goal (SDG) 7: Affordable and Clean Energy, by promoting efficient and transparent management of the electric grid. This research also aligns with the AI & Data Science research cluster, by focusing on explainable AI applications in the realm of energy systems, decision support systems, and behavioral analytics. By bridging the gap between prediction and interpretability, this research aims to empower energy operators and policymakers in Sri Lanka with valuable insights towards promoting energy sustainability.

1.2 Literature Review

1.2.1 Electricity Demand Forecasting and Sri Lanka

While there is still limited research done on electricity demand forecasting in Sri Lanka compared to more studied markets like the U.S. and European grids, existing works mostly concentrate on short-term load forecasting using statistical methods like ARIMA and Multiple Linear Regression (MLR), which, though yielding meaningful results, cannot handle complex and nonlinear patterns of consumption [1]. Singh et al. showed that Multiple Linear Regression models can effectively measure and quantify factors like environmental and temporal factors, such as temperature and time of day, on electricity demand. Yet, they cannot easily adjust to the increasingly fluctuating patterns of modern electricity demand, especially in the context of recent trends in consumption and the integration of renewables.

1.2.2 Machine Learning Approaches for Load Forecasting

Several recent studies have also focused on machine learning approaches, such as Random Forest, Gradient Boosting, and Support Vector Machine, for improving the accuracy of load forecasting models for electricity demand [3]. Machine learning models are better at dealing with non-linear relationships and have been proven to perform better in short and medium term load forecasting applications. However, most of these machine learning models are considered "black box" models, i.e., they provide accurate results, but there is no explanation of what features are responsible for determining the electricity demand. This has also reduced the confidence of energy planners and system operators in adopting these models for use in practice.

1.2.3 Deep Learning Models for Time-Series Forecasting

Deep learning models like Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU) have been found to be effective in the identification of temporal relationships in energy data sequences [4]. A comparative study of the performance of ARIMA, Prophet, and LSTM models showed the superiority of the LSTM-based approach in terms of accuracy for short-term load forecast due to the capacity of the models to learn non-linear and dynamic patterns of energy consumption [5]. In addition, the use of hybrid models has been found to enhance the stability of the forecast. However, the models are not transparent, i.e., the key factors affecting the forecast are not identifiable.

1.2.4 Explainable Artificial Intelligence (XAI) in Energy Systems

New techniques for explainable artificial intelligence, called Explainable AI (XAI), have been developed, which include SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations). These were developed due to the lack of transparency associated with modern forecasting models [6]. These new techniques provide users with the ability to understand the reasons behind the model prediction by quantifying the contribution of each input feature to the prediction result. This enables users to understand why the model predicts a specific value for electricity demand. Recent studies using SHAP for energy systems have found that temperature, historical loads, time of day, and weather conditions are among the features with the highest values for predicting electricity demand [7].

1.2.5 Visualization and Decision Support Tools

Today's forecasting systems are becoming more integrated with visualization dashboards to enable the presentation of the forecast results and explanations in an easily interpretable manner. Feature importance plots, SHAP summary plots, and interactive dashboards are useful tools for the operator to analyze the impact of various variables on the overall electricity demand [8]. Visualization is useful for the planners to identify the causes of peak demands, seasonal demands, and the possible impact of external variables like holidays and public gatherings.

1.2.6 Limitations of Existing Systems

Even though there is a significant improvement in forecasting accuracy with the use of machine learning and deep learning models, it is evident that such models do not incorporate explainability and interaction. Most models do not offer real-time and scenario-based information, making it difficult for the forecasting solutions to be useful in decision-making. Furthermore, it is evident that few forecasting solutions combine high prediction accuracy and clear explanation, making it challenging for the decision-makers to trust the forecasting solutions.

1.3 Research Gap

Most of the existing research on AI-based electricity demand forecasting has concentrated on achieving better accuracy using models such as LSTM, ARIMA, gradient boosting, and other machine learning algorithms [1][2][4][6][9]. Even though these models achieve satisfactory accuracy in forecasting, they lack consideration of the causes of variations in demand, which is an essential factor for proper decision-making and management of energy systems. This deficiency of existing models can be grouped under three major categories:

Key Gaps:

1. Behavioral Analysis: The existing models fail to identify daily consumption patterns, anomalies, and correlations of energy consumption with consumer and industrial behaviors [2][7]. This leads to a poor level of understanding of human activities and their influence on changes in demand, thereby restricting behavioral intelligence.
2. Interpretability: The conventional forecasting models used tend to predict point values but do not show explanations for weather, calendar, socio-economic, and industrial load changes [1][4][8]. This restricts the ability of planners and utilities to understand why changes occur.
3. Integrated Intelligence: The conventional models and tools available tend to have specific functionalities, such as predicting and monitoring, but do not have integrated capabilities for behavioral analysis, predictive modeling, and decision support, including explainable AI [3][9][11]. This is necessary for integrated electricity demand management

Technical Gaps:

1. Limited multi-source data integration: In general, the forecasting models used depend mostly on historical patterns of electricity demand and do not take into consideration other external factors like weather conditions, industrial activities, holidays, etc. [1][5][10]. As a result, they do not fully capture the intricacies involved in the demand for electricity.
2. Limitations of existing LSTM/ARIMA models: Even though LSTM and ARIMA models are used for sequential and short-term forecasting, they do not guarantee long-term prediction accuracy necessary for strategic planning and policy formulation [6][10].
3. Underexplored advanced deep learning models: In general, the scope of the existing literature does not cover hybrid models like CNN-LSTM, attention-based LSTM, etc. [5][11]. These models have the ability to capture intricate temporal and spatial relationships involved in the demand for electricity.
4. Scarcity of hybrid and ensemble models: There is a scarcity of research work where the researchers have attempted to combine various models of deep learning. For example, CNN is useful for spatial data, while LSTM is useful for time series data [4][9].
5. Scarcity of probabilistic forecasting: Most of the researchers have only concentrated on the accuracy of the forecast by using metrics like MAE, RMSE, or MSE. However, the uncertainty of the forecast is not well explored [9][10].

2.OBJECTIVE

2.1 Main Objective

The proposed project is to provide a clear, interactive way of forecasting electricity demand in Sri Lanka. This will be accomplished by producing both short-term and long-term predictions with the combination of machine learning and deep learning models to give an accurate prediction of future electricity demand. The researchers will provide the end users, energy planners; the ability to understand, trust and use the predictive output by providing a dashboard and allow for scenario analyses.

2.2 Specific Objectives

- Designing and creating precise forecasting models for short-term electricity requirements using ARIMA, Random Forest, XGBoost and LSTM/GRU architectures that account for both nonlinear temporal relationships and environmental factors (temperature, humidity, rainfall and time of day) affecting electricity demand.
- Adding an Explainable AI component (USE SHAP & LIME) to each of the four forecasting models to provide a global and local variable importance explanation for every forecast made, thus increasing access & improving model transparency.
- Building an interactive dashboard that will present the output of each forecasting model (planned vs actual loads), how much each of the variables contributed to those loads, peak vs low levels of demand from each of the four models, and allow for scenario simulation so that planners have access to the best available information.
- Including a way for users to alter the external variable input (e.g. temperature, rainfall, etc.) and see what effect these changes had on each model's forecasted demand.
- Optimizing system performance for real-time inference and analysis with low latency and smooth display rendering on standard desktop/web platforms.
- Verifying forecasting based on back-testing against historical data, evaluating forecasting system performance metrics (MAE/RMSE), and collecting feedback from energy planners on usability to establish whether the information provided by each forecasting model is actionable.
- Providing modular integration to existing smart grid systems using REST APIs as a means for connecting to other energy management or decision-support tools.

3.METHODOLOGY

The development of an XAI module which will provide energy planners with transparent and actionable insights into the rationale behind the system's prediction of electricity demand will be developed through this DSR based approach. The DSR methodology allows iterative design, development, and evaluation of technologies, such as XAI, according to established principles of design research (research through design). Quantitatively assessing the reliability of explanations, the accuracy of feature importance rankings, and how well users understand the visual representations of data will provide the basis for evaluation of the XAI module.

3.1. Pre-trained forecast model, data input

Component 3 uses outputs from component 2 (historical electric/weather data database for Sri Lanka) and is a combination of pre-processed data and trained forecasting models. The input features of this module are as follows: temporal information (hour, day, month), weather variables (temperature, humidity, rainfall), event and lagged load value. The module uses XGBoost, Random forest, LSTM or GRU training models with special priority to XGBoost due to its very good capability in using SHAP for determining outputs/interpretation of the prediction. Therefore, no raw data gathering is needed because this module utilizes prepared (cleaned up) data and pre-trained models to generate an explanation about what has caused the different outputs.

3.2.Explainability Methods

The module utilizes explainable artificial intelligence (XAI) methods to calculate the impact of individual input variables on forecasted electric energy usage (demand). The main method used to provide both global and local interpretability is called SHAP (SHapley Additive exPlanations). Other optional methods are LIME, to give interpersonal level explanations, mean decrease accuracy for cross-validation, and partial dependence plots to visually display input variable effects. All data analysis will be conducted using Python libraries like pandas, numpy, scikit-learn, and xgboost.

3.3.Global and Local Explanations

There are two different kinds of explanations being produced. Global explanations find points within the whole dataset which are the most relevant to each feature (hour of the day, temperature, previous load, day of the week, and rainfall) using things like SHAP summary plots or feature importance charts. Local explanations interpret single predictions and show why there is a significantly high or low forecast at that moment in time; for instance, if you look at March 10, 2025 at 6 pm you would see High temp & night time will likely have a load increase, while rainfall would reduce forecasted load slightly. This dual approach provides Planners with the ability to see the big picture and get into the detail.

3.4.Special Case Analysis

This module describes notable prediction types (highest-demand day, lowest-demand day, more common anomaly type and public event day). Each scenario will include an analysis and visualization of feature contributions that represent specific input features that contribute to extreme or unusual electric power consumption, thereby helping planners understand what caused these types of extremes.

3.5.Visualization and Dashboard Development

Data is displayed visually via an interactive dashboard developed for energy planners. Examples of data visualizations include SHAP summary plots, feature importance bar charts, as well as force/waterfall plots. Written explanations provide a clear narrative, such as “the temperature and hour of the day were the most influential contributors to today’s peak load.” In addition, users may use the interactivity of the dashboard to choose a specific date, time or scenario to view both local and global explanations, which will make any insights gained from the database actionable and understandable.

3.6.Backend and Frontend Implementation

Using Python and either Flask or FastAPI, the backend serves both computation of explanatory data (such as SHAP values) and display of features that contributed most to each prediction. A React application (or HTML/JavaScript) is used for the frontend with a responsive and user-friendly dashboard that visualizes all of this information so that management can view interpretable predictions in real time without retraining models.

3.7.Validation Strategy

Validation is the process of determining whether the module is providing accurate, reliable, and understandable explanations. Quantitative evaluation involves measuring explanation fidelity and stability by comparing the outputs of SHAP and LIME to permutation importance and model sensitivity. User evaluation consists of energy planners testing the dashboard by completing structured tasks and engaging in scenario-based exercises while answering structured questionnaires regarding clarity, trust and support for decision-making. The feedback from user evaluations is applied toward improving visualizations and explanation algorithms in successive iterations.

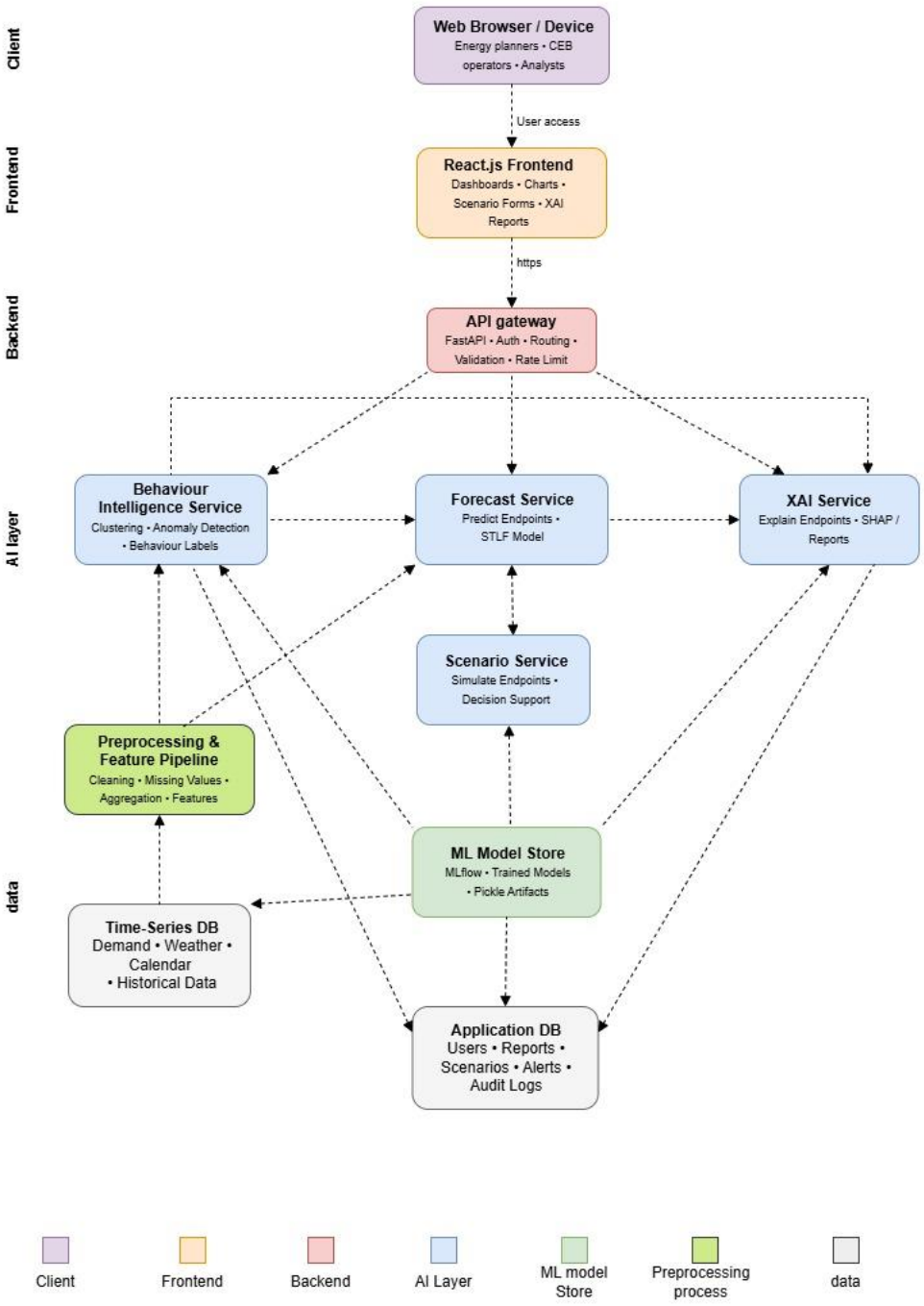


Figure 0-1 System Architecture Diagram

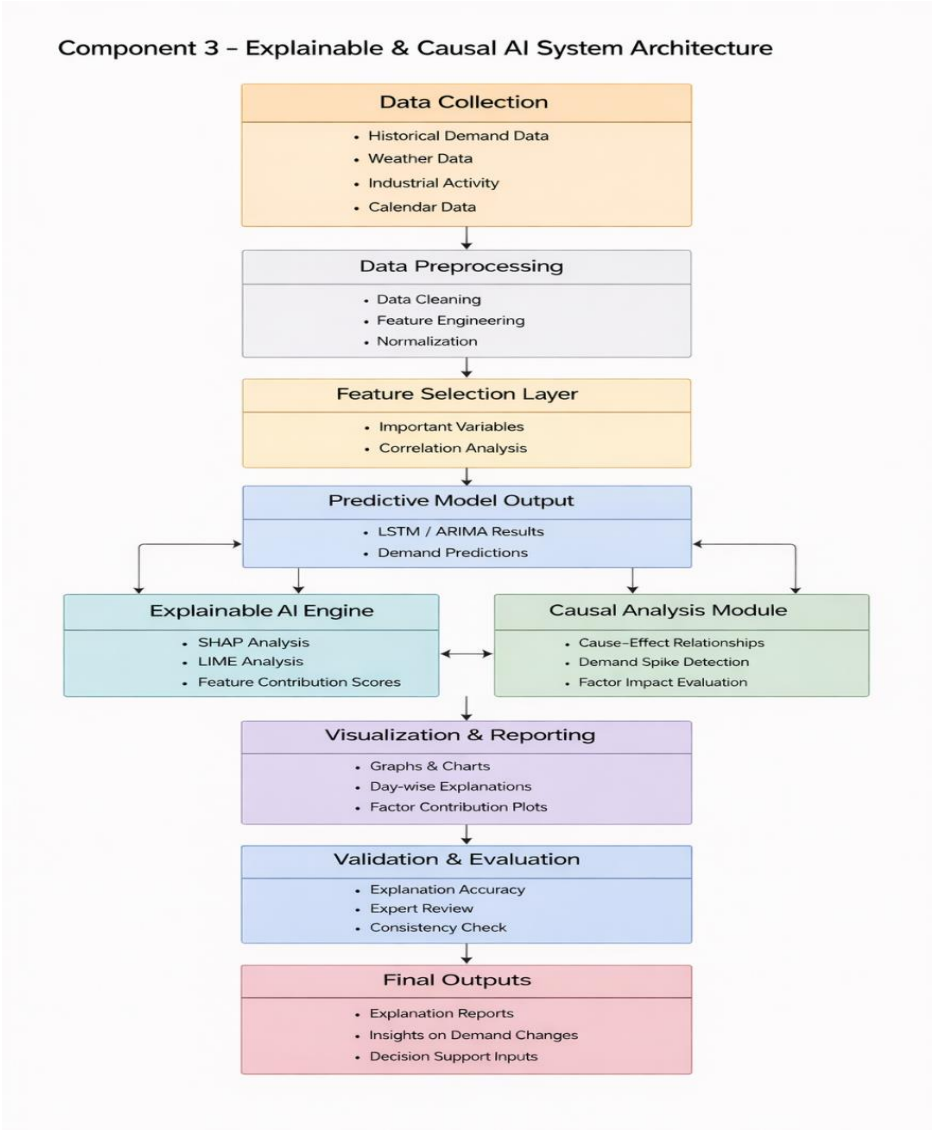


Figure 0-2 Component Architecture Diagram

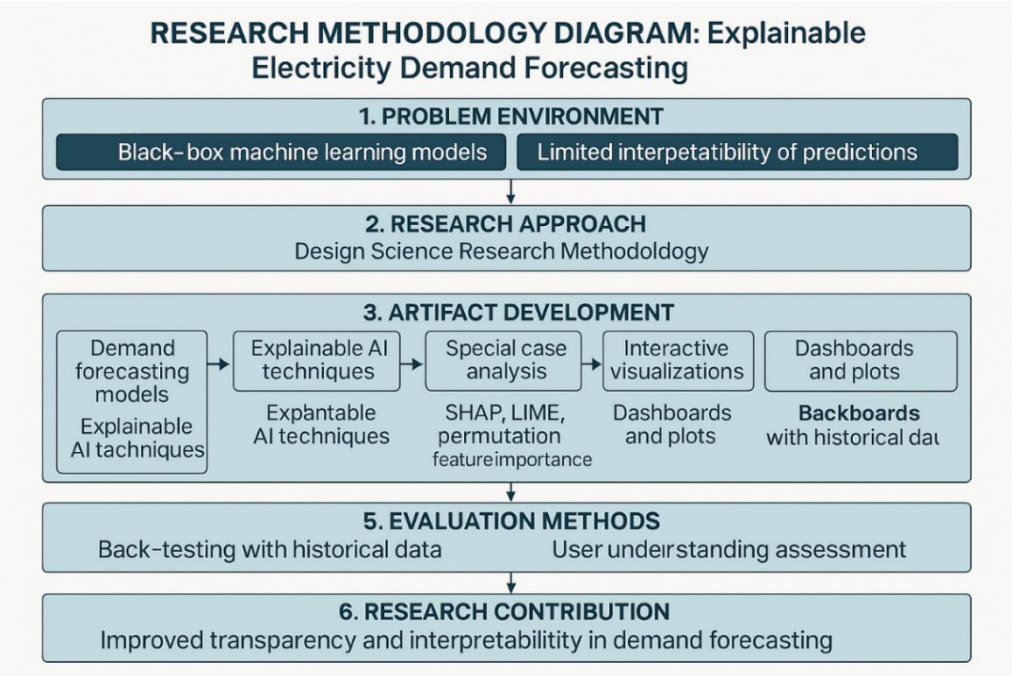


Figure 0-3 Research Methonology Diagram

3.8.Technology Stack

Table 1Technology Stack Summary

Category	Technology	Justification
Forecast Models	XGBoost / Random Forest / LSTM / GRU	Accurate electricity prediction; XGBoost preferred for SHAP explainability
Explainability	SHAP / LIME / Permutation Importance	Produces local and global feature importance, interpretable visualizations
Data Handling	Python / Pandas / NumPy / scikit-learn	Efficient preprocessing and manipulation of large datasets
Visualization	Matplotlib / Plotly	Interactive charts, force plots, and dashboards for decision support
Backend API	Flask / FastAPI	Serves explanation computations to frontend in real time
Frontend	React / HTML / JavaScript	Interactive, user-friendly dashboards for planners
Scenario Analysis	Python + Pandas	Analyze peak/low/anomaly days and generate textual explanations

4.HIGH-LEVEL SYSTEM ARCHITECTURE

The Explainable AI (XAI) module is the interpretability component of the overall electricity demand forecasting system. As a subcomponent of the overall electricity demand forecasting system, the XAI module operates as an interface between the pre-trained forecasting models in Component 2 and the pre-processed historical electricity and weather datasets by producing global and local explanations that are then presented on an interactive dashboard to energy planners and grid operators for use in making decisions regarding energy use. The Python backend (using Flask or FastAPI) will serve as an API to provide real-time access to SHAP/LIME computations, visualizations and textual explanations.

4.1 Module Overview

The XAI module is structured across three main stages:

- Stage of input: Pre-processed historical weather dataset and electrical datasets (as well as previously trained predictive models XGBoost, Random Forest, LSTM, and GRU) are input.
- Stage of explanation: ASHS (primary) and LIME or permutation importance (optional) calculate global and local feature contributions to the prediction (actual value) as well as special cases (peak, low, anomalies, and public event days).
- Stage of visualization: Visualizations (SHAP summary plots and force plots, waterfall plots, bar charts) and explanations (text based) of the predictions are used to generate dynamic dashboards.

4.2 Data Flow

1. The user chooses a day/time or scenario for analyzing electricity demand through the dashboard.
2. The backend loads the pre-processed dataset and the associated forecasting model.
3. The model computes the shapley values for either global (using all datasets) or local (individual dataset) explanations.
4. The feature importance is ranked according to inputs and any special-case insights generated.
5. Visual representations (summary plots, bar charts, force/waterfall plots) and text explanations are created.
6. The user's dashboard displays the visual representations and text to the end user; with the ability to select and/or filter datasets by input feature, date, or simulation scenario.
7. All of the actions performed by the user (e.g. simulation of a scenario) are logged and sent to back-end datastore for analysis and improvement.

4.3 Backend Architecture

The following is a summary of the RESTful Web Service API calls made by the dashboard on the front end, which is based on a backend written in either Python Flask or Python FastAPI:

- POST to "/API/explain/global" — provides a global feature importance summary for the selected model.
- POST to "/API/explain/local" — provides a local explanation for a given prediction timestamp.
- POST to "/API/explain/special-case" — returns information about days when there is peak, low, anomaly, or public event activity.
- GET to "/API/features" — returns metadata for all of the features used in the model.

All endpoints are secured by using HTTPS/TLS and a token-based authentication mechanism. The backend can be easily scaled horizontally through containerization or through deploying to a Cloud platform, allowing for simultaneous requests (converging explanations) by users.

4.4 Integration with Other Components

The module provides explainable output for analysis that is provided to the Electricity Demand Behavior Intelligence module (Thanujan); this helps in interpreting detected electrical demand patterns and anomalies; for the Adaptive Forecasting & Prediction System (Janahan) this helps to give transparent justification for predicted electricity demand values; and it is used by the Scenario Analysis & Decision Support System (Dewmini) as input for the support of "What-if" scenario simulations and risk assessment of electricity demand planning. All components use the same unified electricity demand data as well as weather and calendar data, which is stored in a single centralized database; this ensures that there is consistency of data used between each module for the analysis of behavior, forecasting, explainability, and scenario simulation. All inter-component communication and data exchange is performed through the shared backend processing layer; this allows for complete and consistent integration among, and flow of information between, the different modules in the system.

4.5 Scalability and Privacy

The design of the system architecture guarantees that all computations will occur on the backend server to keep confidential historical electric and weather data safe (e.g., the raw datasets will not be exposed to the frontend dashboard). Only aggregated output (including SHAP or LIME values, visualizations, and textual explanations) will be transmitted to the dashboard, thus minimizing the risks related to privacy. The modularity of the XAI engine enables continued updates and/or extensions (such as adding LIME explanations, adding new forecasting models, and/or enhancing visualisation capabilities) without affecting the forecasting module or the frontend interface, making this system scalable and maintainable.

5.USER REQUIREMENTS

5.1Stakeholder Identification

- **Primary Users:** Electricity Planning/Systems Operators who will be reviewing electricity demand forecasts.
- **Secondary Users:** Staff at Utilities and Grid Managers will use these dashboards to make operational decisions.
- **Tertiary Users:** Researchers & Analysts; Policy Makers reviewing/useable interpretable electricity demand metrics.
- **System Admin:** SLIIT researchers and any future maintainers of the XAI module.

5.2Functional Requirements

Table 2 Functional Requirements

FR	Category	Requirement
FR1	Data Input	System shall accept preprocessed historical electricity and weather datasets including temporal, weather, event, and lag features.
FR2	Forecast Model Integration	System shall integrate outputs from pre-trained forecasting models (XGBoost, Random Forest, LSTM, GRU) for explainability analysis.
FR3	Explainability Methods	System shall generate both global and local explanations using SHAP; optionally LIME and permutation importance.
FR4	Global Explanation	System shall identify and rank most influential features across the dataset and visualize results via summary plots and bar charts.
FR5	Local Explanation	System shall explain individual predictions, showing positive or negative contributions of each feature.
FR6	Special Case Analysis	System shall analyze and explain peak-demand, low-demand, anomaly, and public event days with textual and visual insights.
FR7	Visualization Dashboard	System shall present explanations through interactive visualizations including SHAP summary plots, force/waterfall plots, and textual narratives.
FR8	User Interaction	System shall allow users to select specific dates, hours, or scenarios to view detailed explanations.
FR9	Backend API	System shall provide REST endpoints via Flask/FastAPI for computation and retrieval of explanation results.
FR10	Integration	System shall integrate explanation outputs into operational and planning dashboards without disrupting existing forecasting models.

5.3 Non-Functional Requirements

Table 3 Non Functional Requirements

Category	Requirement
Performance	SHAP/LIME computations served to frontend within 500ms; dashboard visualizations rendered interactively without lag.
Accuracy	Feature importance rankings must be consistent and explain at least 90% of prediction variance; local explanations should accurately reflect model contribution.
Usability	Dashboard must be intuitive for energy planners; textual and visual explanations should be understandable without technical AI knowledge.
Reliability	System shall handle large datasets without crashes; backend failures should fallback gracefully to cached explanation results.
Privacy	Raw historical datasets must not be transmitted to frontend; only aggregated or anonymized SHAP/LIME values, visualizations, and textual insights are shared.
Compatibility	Dashboard accessible on modern web browsers; backend and XAI computations compatible with server infrastructure supporting Python 3.9+ and required libraries.
Security	All API communications secured with TLS 1.3; user access controlled with JWT authentication; sensitive dataset access restricted to authorized administrators.

6.COMMERCIALIZATION PLAN

1. **Target Market:** The primary target market consists of national energy utilities and grid operators that are located in developing countries, with a focus on the South Asian region and beginning in Sri Lanka. Secondary markets include energy regulators, large-scale industrial zones, renewable energy developers and smart city planners. Customer persona: The Decision Analyst Job title: Energy Analyst or Planning Engineer at a utility organization (e.g., Ceylon Electricity Board). Pain points: Difficulty understanding why there is fluctuation in electricity demand; lack of transparency in AI predictions; cannot justify demands to management due to spikes in demand. Goals: To be able to understand demand patterns & behaviors clearly, identify major factors influencing those behaviors, and make explainable, data-driven decisions.
2. **Value Proposition :**The platform offers an Explainable AI layer instead of merely being a predictive tool. The system changes the perception of AI from being a black box into an explainable system; with the system providing insight into the reasons for demand fluctuations due to things like weather, holidays, and pricing using analysis methods such as SHAP and LIME. The system will develop Causal Insights; providing an understanding of the cause and effect relationship and the true drivers causing changes in demand. Pro-actively users will have the ability to read and respond to output from the AI model with confidence while making educated operational decisions based on intelligible rationales.
3. **Business to Government (B2G)** : The annual subscription model (Software as a Service - SaaS) provided to national energy utilities; Business to Business (B2B) -Licensing of the Explainable AI module to private energy companies and industrial consumers; Consultancy and Integration Fee- One Time Revenue from the integration of the Explainable AI module with existing analytics systems, dashboards or energy management platforms.
4. **Development Costs:** The cost of developing Explainable AI methods (SHAP & LIME) which are used for interpreting Classifier Models, data processing, and cloud computing resources. Data/Purchasing Expense: Cost associated to retrieving, cleaning, and updating electricity, weather, and event data; Operating Expense: Cost of maintaining servers, re-training the models, updating systems and providing technical support so you continue to operate well and reliably.
5. **5. Pricing Based on Value**: The pricing will be decided on the value associated with increased insight and making better decisions in managing energy as a result of this service. The pricing will be a subscription service with a tiered pricing model based on the level of use, number of users and the features accessed within the system. Pricing for pilot programs will be set at an initial cost so that local utilities can conduct a pilot deployment of the service to confirm the value of using the service and demonstrate its value.
6. **Explainability as a Core Feature:** This system is unlike most others, in that it explains how things are predicted/diagnosed, thereby giving us confidence in what is being identified; it can identify not only correlation but also true and identifiable causal linkages between the items driving demand; it allows for visualisation of the demand in an easy-to-use manner by non-technical individuals; and it has been designed to specifically cater to the local environment, i.e., by including local considerations of the characteristics of the weather in the region and the way in which holidays impact consumption within that area.
7. **Algorithm Design:** The proprietary implementation of Explainable AI techniques for analyzing electricity demand. Trade Secrets: Protection against public disclosure of the internal logic used in explanation generation and visualization processes. Software Copyright: Copyright of the source code pertaining to the Explainable AI module and reporting interface.

7.BUDGET AND JUSTIFICATION

The Explainable & Causal AI Platform has access to high performance hardware and cloud infrastructure which means the platform can effectively handle complex time-series data, without any bottlenecks, which will optimize the training of hybrid models (CNN - LSTM and Transformers) for capturing non-linear patterns of electricity demand.Dedicated storage and cloud resources allow the Explainable & Causal AI platform to create interpretations of data on weather, industrial, and calendar variables to produce probabilistic forecasts and quantify uncertainty. Field verification and the transportation of material will be required to validate anomalies by comparing to what actually transpired in the real world through a utility office (CEB) or weather station so that the outcomes of behavioral analysis and scenarios simulation are accurate and actionable.Together, all these costs provide the essential infrastructure which supports the delivery of dependable, interpretable and adaptable electricity demand intelligence to energy planners in Sri Lanka.

Table 4 Budget and Budget Justification

Item	Qty	Unit (LKR)	Total (LKR)	Justification
Laptop / Workstation for Model Development	1	150,000	150,000	Machine learning model analysis, SHAP computations, and dashboard development
Cloud Computing (Google Colab Pro / GPU)	12 months	4,500	54,000	Training and evaluating ML models (XGBoost, LSTM, Random Forest) using GPU resources
Web Hosting / Cloud Server	12 months	2,500	30,000	Hosting backend APIs (Flask/FastAPI) and explanation services
Domain & SSL Certificate	1	5,000	5,000	Secure web access for dashboard and API endpoints
Data Storage & Backup (Cloud Storage)	12 months	1,500	18,000	Storage for electricity demand datasets, weather data, and model outputs
Visualization & Analytics Tools	1	6,000	6,000	Tools/libraries for creating interactive dashboards and visualization reports

Data Collection / Research Transport	3	6,000	18,000	Travel for meetings, dataset collection discussions, and research coordination
Miscellaneous (printing, documentation, internet)	1	10,000	10,000	Report printing, documentation preparation, and research communication
			TOTAL BUDGET:	**LKR 291,000**

8.WORK BREAKDOWN STRUCTURE (WBS)

Table 5 work breakdown structure

WBS	Task	Sub-Tasks	Timeline
1.0	Research & Requirements	Literature review on XAI in energy systems, identification of stakeholders, requirement gathering, dataset assessment, selection of features and models	Jan – Feb 2026
2.0	Data Preparation	Cleaning historical electricity and weather data, generating lag features, event flag encoding, dataset splitting for training/testing	Feb – Mar 2026
3.0	Forecast Model Integration	Import pre-trained models (XGBoost, Random Forest, LSTM, GRU), verify performance metrics, ensure compatibility with SHAP/LIME pipelines	Mar 2026
4.0	Explainability Method Implementation	SHAP integration for global and local explanations, optional LIME implementation, permutation importance calculations, partial dependence plots	Mar – Apr 2026
5.0	Special Case Analysis	Identify highest-demand, lowest-demand, anomaly, and public event days; compute feature contributions; generate scenario-specific insights	Apr 2026
6.0	Visualization & Dashboard	Design interactive dashboard, implement SHAP summary plots, force/waterfall plots, feature importance charts, textual insight	Apr – May 2026

		display, interactive filters	
7.0	Backend Development	Setup Flask/FastAPI backend, serve SHAP/LIME computations, provide REST API endpoints, integrate with frontend dashboard, ensure data privacy and secure access	May 2026
8.0	Frontend Development	Develop React/HTML/JS dashboard interface, interactive visualization components, scenario selection controls, real-time display of explanations	May – Jun 2026
9.0	Validation & Testing	Quantitative evaluation of SHAP/LIME stability and explanation fidelity, user testing with energy planners, scenario-based exercises, feedback incorporation	Jun – Jul 2026
10.0	Documentation & Submission	Technical report writing, dashboard user guide, API documentation, research paper draft, final project submission	Jun – Aug 2026

9.GANTT CHART

Table 6 Gantt Chart

System Development Project Timeline – 2026													
		● Semester 1 (Jan – Jun 2026)						● Semester 2 (Jul – Dec 2026)					
Phase	**Activity**	**Jan**	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
Phase 1 – Research & Design													
Phase 1	1.0 Research Requirements	·	·										
Phase 1	2.0 System Design		·	·									
Phase 2 – Data & Model													
Phase 2	3.0 Dataset Preparation			·	·	·							
Phase 2	4.0 Model Development					·	·	·					
Phase 3 – Implementation													
Phase 3	5.0 Backend Development							·	·	·			
Phase 3	6.0 Frontend Development									·	·		
Phase 3	7.0 System Integration											·	
Phase 4 – Testing & Delivery													
Phase 4	8.0 Testing & Evaluation												·
Phase 4	9.0 Documentation & Final												·
Summary	**Tasks per Semester**	**Semester 1: Tasks 1.0 – 4.0 | Research → Model Development**						**Semester 2: Tasks 4.0 – 9.0 | Development → Delivery**					

10.Referencess List

[1] D. A. Bolstad, U. Cali, M. Kuzlu and U. Halden, *"Day-ahead load forecasting using explainable artificial intelligence,"* in *2022 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT),* New Orleans, LA, USA, 2022.

[2] G. E. Alilu, G. C. Okorie, I. A. Ayoade, U. O. Obonna, O. K. Akinde and C. I. Nwakanma, *"Explainable artificial intelligence model for electricity usage prediction in Nigeria,"* in *2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON),* Ado Ekiti, Nigeria, 2024.

[3] N. Ahmad, Y. Ghadi, M. Adnan and M. Ali, *"Load forecasting techniques for power system: research challenges and survey," IEEE Access*, vol. 10, 2022.

[4] T. Hachiya and K. Kamei, *"Prediction of power demand and factor identification using sparse modeling AI with ensemble learning for improvement of accuracy,"* in *2024 Joint International Conference on Soft Computing and Intelligent Systems and International Symposium on Advanced Intelligent Systems (SCIS&ISIS),* Himeji, Japan, 2024.

[5] P. Joshi, J. Bhardwaj, S. Størdal, M. Kolhe and E. Haugom, *"Analysis of feature engineering in deep learning architectures for dynamic electricity price forecasting,"* in *2025 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA),* Antalya, Turkiye, 2025.

[6] D. P. A. et al., *"Electrical load forecasting using ARIMA, Prophet and LSTM networks," International Journal of Electrical and Electronics Research (IJEER)*, 2021.

[7] N. Sakkas et al., *"Explainable approaches for forecasting building electricity consumption," MDPI,* Basel, Switzerland, 2023.

[8] U. Pati and K. D. Mistry, *"Explainable artificial intelligence approach for demand-side management in a 1-phase multi-type consumer base: enhancing efficiency and transparency," IEEE Transactions on Emerging Topics in Computational Intelligence,* vol. 9, no. 4, 2024.

[9] S. S. Shyam et al., *"Electrical load forecasting with anomaly detection and explainability," IJRASET – International Journal for Research in Applied Science & Engineering Technology,* Pottapalayam, Sivagangai, India, 2025.

[10] Y. Fu et al., *"Masked multi-step probabilistic forecasting for short to mid term electricity demand," arXiv,* Ithaca, New York, USA, 2023.

[11] C. Yan and H. Ruan, *"Power load and price forecasting based on spatiotemporal decoupling and explainable artificial intelligence interaction perception,"* in *2025 5th International Conference on Electrical Engineering and Mechatronics Technology (ICEEMT),* Shenzhen, China, 2025.

11.APPENDICES

Appendix A: Risk Analysis

Table 7 Risk Analysis

Risk	Likelihood	Impact
Insufficient historical electricity and weather data for Sri Lanka	Medium	High
SHAP/LIME explanations misrepresenting factor contributions	Medium	High
LSTM/ARIMA model underperforming on short- or long-term demand	Medium	High
Scenario simulations producing unrealistic outputs	Medium	Medium
Computational delays in generating forecasts and explanations	Low	Medium
Integration issues between prediction, explainable AI, and scenario modules	Medium	Medium
Misinterpretation of visualized causal insights by end-users	Medium	Medium
Missing or inconsistent calendar/holiday data affecting model accuracy	Low	Medium

Appendix B: Ethical Considerations

- This project will collect aggregated and de-identified electricity & weather data from publicly available databases; therefore no private or sensitive data will be collected or used.
- The handling of all data will comply with appropriate usage and responsible practices to ensure that transparency and integrity are maintained during model development/creation as well as during the analysis of the data.
- Any and all outputs and visualizations generated by the project will not show any identifying information about individual electricity consumers; therefore, confidentiality and privacy will also be maintained.
- Approval for the project from the SLIIT Ethics Review Committee will be pursued in order to ensure compliance with SLIIT's guidelines.
- The research will also conform with the Personal Data Protection Act No. 9 of 2022. This will guarantee the lawful and responsible use of data.
- Scenario simulations and predictive models used in this project will be developed in a manner that will eliminate the potential for misuse or distorted interpretations; all results will have sufficient explanations provided to ensure clarity for any potential user.

- All project output will reflect the cultural, social, and energy usage contexts to ensure that the outputs of the study are appropriate and relevant for all stakeholders who are involved in electricity planning in Sri Lanka.

Appendix C: Technology and Framework Stack

- The following technologies are used in the implementation of the Explainable AI module for electricity demand forecasting:
- Python (core programming language - 3.11+) used to pre-process the data and perform the feature engineering to train and explain models.
- pandas and NumPy provide data handling capabilities and numerical computations for time-series demand data and derived features.
- scikit-learn provides machine learning utilities for pre-processing, evaluating models and calculating feature importance.
- XGBoost, Random Forest, LSTM, and GRU can be used to forecast electricity demand, with the recommended choice being XGBoost due to its SHAP explainability.
- SHAP (SHapley Additive exPlanation) is the primary explainable AI tool, which provides global and local feature importance, as well as force and summary plots.
- LIME is an optional tool for local interpreting of a single prediction.
- Permutation Importance and Partial Dependence Plots provide additional explainability methods for understanding the impact of features.
- Matplotlib, Plotly, and Seaborn are visualization libraries that can be used to generate charts of feature importance, waterfall plots, and interactive dashboards.
- Flask and FastAPI are backend API frameworks that are used to serve model predictions and provide explanations through a web interface.
- React, HTML, and JavaScript are frontend frameworks that are used to present interactive visualizations of SHAP and LIME outputs, as well as an explanation dashboard.
- SQLite and PostgreSQL are databases for storing historical demand data, weather features, and model output.
- Git + GitHub (with GitHub Actions) provide version control and CI/CD pipelines for reproducibility of development, testing, and deployment.
- Jupyter Notebook and Visual Studio Code are development environments for experimentation with models, analysis of explainability, and creation of visualizations.

Appendix D: Preliminary UI Mockups – Explainable Electricity Demand Screen

The Explainable AI electricity demand analysis screen is designed with the following key elements:

- At the top of the dashboard you will see a section for Predicted Electric Demand for the selected time period (i.e. 2100 kW).
- A feature importance bar chart shows you which factors are most influential in the prediction of electric demand; these include temperature, hour of day, total rainfall and day of week.
- A SHAP summary visualization will show you what features influence the predictions the most overall.
- A Local Explanations panel contains a graphical serial representation (i.e. force or waterfall) explaining the reason for a high or low predicted electric demand.
- An interactive time selection allows for selecting any date and time to view electric demand predictions.

- The Peak Electric Demand (high consumption) explanations shows the factors contributing to high electric consumption periods.
- The Low Electric Demand (low consumption) explanation will show the reason(s) for lower than expected electric consumption.
- A text explanation box provides some simple examples of insights that can be gained from the analysis, such as: "Temperature and evening hour increased electric demand during this time period."
- Scenario Analysis controls allow the user to modify variables (e.g. temperature and total rain fall) and see how these variables affect electric demand predictions.